Latent Trajectories of Screen Time and Their Predictors in School-Aged Children: A Six-Year Longitudinal Analysis

Sunmi Cho; Saebom Jeon; Sang-ha Lee; Ji-hye Song; Yunmi Shin
Psychiatry Investigation · 2026 · PMC13571864 · DOI: 10.30773/pi.2025.0364

Resumo (Abstract)

Objective Children's screen time has increased rapidly with the rise of digital technology and was further accelerated by the COVID-19 pandemic, heightening concerns about developmental impacts. This study investigates six-year trajectories of children's screen time, identifies usage profiles, and examines key social and behavioral factors influencing these patterns during and after the pandemic. Methods We analyzed data from 313 children participating in a prospective South Korean cohort study (2018–2023). Caregivers reported weekday and weekend screen time on smartphones, tablets, and personal computers (PCs). Joint latent class profile analysis (JLCPA) identified trajectories, and multinomial logistic regression tested the influence of child sex, age, parental education, household income, and Child Behavior Checklist (CBCL) scores. Results Screen time increased across devices, particularly smartphones and PCs. JLCPA revealed four distinct trajectories: low-risk (27.2%), mid-risk (38.4%), risk-increasing (15.3%), and high-risk (19.1%). Membership in the high-risk trajectory increased during and after the COVID-19 pandemic. Significant differences were observed across trajectories, with lower maternal education and male sex emerging as the most robust predictors of higher-risk membership. Children whose mothers had lower educational attainment were up to six times more likely to belong to the high-risk trajectory. Older age and higher caregiver-reported CBCL scores were also associated with increased odds of membership in higher-risk trajectories. Conclusion Children's screen time behaviors are diverse and shaped by developmental, familial, and socioeconomic factors, with notable pandemic-related changes. Tailored interventions and continued research are essential to promote healthy digital habits and mitigate potential risks.

INTRODUCTION

Emerging digital technologies, including mobile and interactive screen media, are of time viewing screen media [1 , 2]. This increase in screen time is clinically significant because it is associated with brain development, which has important consequences for cognitive and motor development, learning and memory, emotional regulation, and overall health [3 - 7]. Children's brain development is significantly influenced by early environeISSN 1976-3026 OPEN ACCESS mental experiences and has been found to be particularly vulnerable to the effects of screen exposure [8 , 9]. As evidence emerges that screen time negatively affects child development, several longitudinal and cross-sectional studies have investigated the risk factors that influence screen time [2 , 10 - 13]. Research has consistently shown that parental factors, particularly maternal factors, are strongly associated with children's screen time. Among demographic factors, parental education level, household income, and minority status are significantly correlated with screen time [14 - 16]. Furthermore, demographic factors, such as education, income, and mental health, ultimately influence how parents manage their children's screen time [17]. Parental self-efficacy in controlling screen time [18], descriptive norms, and strong commitment to limiting screen time11 likely require consistent and sustained management. As smart devices become more diverse and their range of use expands, large-scale longitudinal studies involving various age groups in different environments

are increasingly needed to support the conclu-sion that screen time leads to negative outcomes across most areas [19 , 20].

The COVID-19 pandemic, which significantly altered people's daily lives worldwide, occurred during the study period. The COVID-19 pandemic led to laws and policies including national school closures, lockdown or shelter-in-place restrictions, and social-distancing recommendations that may have resulted in higher overall screen time among children and adolescents during that time [16]. Extended school closures likely resulted in screen time patterns that differed substantially from typical trends. The increase in screen time during this period was demonstrated to significantly exceed what could be attributed to normative developmental changes and secular trends [21].

This study aimed to identify key predictors of changes in children's smart device usage patterns as they develop. This required a comprehensive approach incorporating various factors, reflecting the types of devices used, differences between weekday and weekend use, effects of age and time on usage patterns, and influence of family characteristics. We achieved this by conducting a six-year longitudinal cohort study tracking and analyzing smartphone, tablet, and personal computers (PCs) usage. This approach allowed us to capture the effects of screen time and reflect diverse aspects of screen use, including differences in patterns between weekdays and weekends and across devices. Furthermore, we classified children into subpopulations (i.e., distinct profiles) based on their screen time patterns and identified key predictors that influence these profiles.

METHODS

Participants

The Kids Cohort for Understanding Internet Addiction Risk Factors in Early Childhood (K-CURE) study was the first long-term observational prospective cohort study to investigate the causes and effects of internet-related diseases and disorders in South Korea. The present study was conducted as a part of the K-CURE study and used data from Waves 4 to 9.

Caregivers (mostly mothers) were invited to participate in a self-administered survey on their children's use of various media platforms. A total of 350 children were initially recruited at Wave 4, the baseline for the present analysis, through community centers in Suwon, Seongnam, and Goyang. These cities are predominantly urban and suburban areas in Gyeonggi Province, the most populous province in South Korea and part of the Seoul metropolitan area; therefore, the sample primarily represents urban and suburban families rather than rural populations. Wave 4 was conducted between October and November 2018, and subsequent waves were assessed annually during the same months (October–November) from 2019 to 2023. For analytic purposes, Waves 4–5 were classified as the pre-COVID-19 period, Waves 6–7 as the COVID-19 period, and Waves 8–9 as the post-COVID-19 period. Although 350 children were initially recruited at Wave 4, the main longitudinal trajectory analysis was based on a balanced panel of 313 participants who provided complete data across Waves 4–7. After Wave 7, the sample size decreased because of natural attrition, with 270 and 257 participants remaining at Waves 8 and 9, respectively. Missing data in Waves 8 and 9 were handled using full information maximum likelihood estimation under the missing-at-random assumption.

Measures

Screen time

Caregivers reported their child's screen time separately for three smart devices (smartphones, PCs, and tablets) on weekdays and weekends. The original response options for average daily media use were "never," "less than 1 hour," "1 to less than 2 hours," "2 to less than 3 hours," "3 to less than 4 hours," and "more than 4 hours." Because adjacent categories could be ambiguous at their boundaries (e.g., a child using a device for exactly 2 hours per day could reasonably be classified as either "1 to 2 hours" or "2 to 3 hours"), and because some categories showed sparse frequencies and category-ordering issues in the Rasch model, the original responses were recoded into a 3-point scale: 0 hours, less than 2 hours, and 2 hours or more. The Rasch model supported the adequacy of this recoding in distinguishing ordered levels of screen time; therefore, the 3-point scale was used in all subsequent analyses.

Predictors

Sociodemographic characteristics were assessed based on self-reports. Categorical variables included the child's sex (male or female) and parental education level for both mothers and fathers (college or higher vs. high school or less). Age (in months) was modeled as a continuous covariate and household income as an ordinal covariate, both treated as timevarying and updated at each wave.

We used the Child Behavior Checklist (CBCL) to assess children's behavioral and emotional problems. The CBCL is a widely validated instrument that evaluates a broad range of behavioral and emotional difficulties, including anxiety/depression, withdrawal, and attention problems [22]. It yields standardized T-scores (mean=50, standard deviation [SD]=10), with higher scores indicating greater behavioral and emotional problems. In general, T-scores below 60 are considered within the normal range, scores of 60–63 are considered borderline, and scores of 64 or higher are considered within the clinical range. The Korean version of the CBCL has also been validated for use with South Korean children [23].

Statistical analysis

In this longitudinal study, after preliminary analyses such as descriptive statistics and exploratory data analysis on key variables, we used joint latent class profile analysis (JLCPA) to identify latent class (LC) trajectories of children's smart device screen time patterns over six years. JLCPA is an advanced statistical method designed to classify individuals into homogeneous subpopulations based on their trajectories of multivariate categorical responses over time [24 - 27]. This method allowed us to identify distinct subgroups of children whose screen time patterns varied across devices (e.g., smartphones, tablets, and PCs) and contexts (e.g., weekdays versus weekends) follow similar trajectories, providing a detailed view of how these patterns evolve over time.

To determine the optimal number of LCs and joint latent profiles in the JLCPA models, we estimated models with increasing numbers of classes and compared model fit across solutions using log-likelihood, Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and bootstrap likelihood ratio test (BLRT) p-values [25 , 28 , 29]. Because conventional chi-square-based fit statistics are not well suited for LC models, absolute model fit was additionally evaluated using a parametric bootstrap procedure, with a bootstrap p-value greater than 0.05 indicating no substantial discrepancy between the observed and model-implied data. Relative model fit was assessed using AIC and BIC, with lower values indicating better fit; AIC generally favors more complex models, whereas BIC imposes a stronger penalty for model complexity. The final model was selected on the basis of statistical fit indices, parsimony, theoretical relevance, and interpretability.

Once the latent structure was identified, multinomial logistic regression was conducted with Profile 1 as the reference category to examine predictors associated with membership in distinct screen time profiles. Sex and parental education (high school or less vs. college degree or higher) were entered as categorical predictors, with “male” and “college degree or higher” serving as the respective reference groups. Household income was treated as an ordinal variable reflecting increasing income levels and was modeled as a continuous predictor. Age (in months) and CBCL scores were included as continuous predictors. Odds ratios (ORs) reflect the multiplicative change in the odds of belonging to a given profile relative to the reference profile. For continuous predictors (age and CBCL scores), ORs represent the change in odds associated with a one-unit increase. For income, ORs reflect the change in odds associated with a one-category increase along the ordinal scale. For categorical predictors, ORs compare each category against its designated reference group.

Ethics

This study received approval from the Institutional Review Board (IRB) at the Ajou University School of Medicine, Suwon City, South Korea (IRB No. AJIRB-SBR-SUR-22-285). Informed consent was obtained from all participants at enrollment. All participation was voluntary, and informed consent was meticulously obtained from both legal guardians and the children themselves.

RESULTS

Sample characteristics

Table 1 presents the demographic and behavioral characteristics of the sample for Waves 4 and 9, selected to represent key time points from the six waves conducted over six years while conserving space. Wave 4 included 313 children, of whom 49.2% were female, with a mean age of 84.3 months (approximately 7 years old). Parental education was predominantly at the college level or higher, with 90.1% of fathers and 90.8% of mothers holding a college degree or higher. Household income most commonly fell within the 4M–6M KRW range (approximately USD 3,100–4,600), accounting for 41.5% of the sample. The mean CBCL score was 47.3 (SD=10.4), which falls within the normal range of the standardized Tscore distribution. By Wave 9, 257 participants remained in the sample. Despite attrition, the sex distribution and parental education levels remained largely stable across waves. Household income increased over time, with a greater proportion of families reporting monthly incomes above 6M KRW (approximately USD 4,600). The mean CBCL score remained stable at 47.5 in Wave 9, suggesting consistently low levels of behavioral and emotional problems throughout the follow-up period.

Screen time behaviors on smart devices during weekdays and weekends

Table 2 shows increases in screen time for smartphones, PCs, and tablets from Waves 4 to 9 on both weekdays and weekends. Smartphone use increased significantly, with the percentage of children reporting no use dropping from 40.6% to 6.6% on weekdays and from 34.8% to 8.6% on weekends. Similarly, PC use increased, with those reporting no use decreasing from 75.7% to 47.9% on weekdays. Tablet use remained relatively stable, although non-use decreased slightly and use for less than 2 hours increased. Overall, screen time increased across all devices, especially on weekends and for smartphones.

Three LCs for children's weekday and weekend screen time were identified (Figure 1A) based on relative and absolute fit indices (AIC, BIC, log-likelihood, and bootstrap p-values), as this solution showed the lowest AIC, competitive BIC, the highest log-likelihood, and an adequate bootstrap p-value (>0.05). For weekdays, the first LC showed the highest proportion of non-use across all devices, whereas the second class generally used devices for less than 2 hours. The third class used devices, primarily smartphones (denoted as "SMTP"), for more than 2 hours and showed higher PC use than the other LCs. These classes were named "no-use," "light-use," and "SMTP heavyuse," respectively. Similar patterns were observed for weekends, with higher smartphone use than on weekdays.

Four joint latent classes, combining weekday and weekend LC patterns, were also identified (Figure 1B), yielding a wellfitting and substantively interpretable solution: "No–No," "No– Low," "Low–Low," and "High–High." The No–No group showed no use on either weekdays or weekends, and the "No–Low" group reported no use on weekdays and light use on weekends. The Low–Low group consistently displayed light use in both weekdays and weekends, whereas the High–High group exhibited heavy use on both weekdays and weekends, especially on smartphones. This classification helps identify device usage patterns, with the High–High group demonstrating a potential need for targeted interventions to address consistently high smartphone use.

Transitions in joint latent classes across waves during and after the COVID-19 period

Figure 2 illustrates the distribution and transition of joint latent classes for school-aged children's screen time behavior across six waves (Waves 4–9, 2018–2023), including key changes during (2020–2021) and after (2022–2023) the COVID-19 period. The No–No group (no use on weekdays or weekends) showed a decreasing trend over time, although an influx from the No–Low group (no weekday use, light weekend use) caused a temporary increase during the COVID-19 period. During the COVID-19 period, many in the No–Low group transitioned to either the No–No or Low–Low (light use on weekdays and weekends) group. Post-COVID, the proportion of children in the High–High group (heavy use on weekdays and weekends) increased significantly.

Four profiles in school-aged children identified using JLCPA

Figure 3 presents four area charts, each depicting different profiles of children's screen time over six waves (Waves 4–9, 2018–2023), categorized into four joint latent classes: No–No, No–Low, Low–Low, and High–High.

Profile 1 (blue) is generally dominated by the No–No pattern, with a brief No–Low phase that disappears during the COVID-19 period, followed by an increase in Low–Low post-COVID. Profile 2 (green) begins with a prominent No–Low pattern, gradually transitions to Low–Low, and shows a small presence of High–High in the final wave. Profile 3 (yellow) starts with No–Low, shifts to predominantly Low–Low during the COVID-19 period, and displays a notable rise in High– High use post-COVID. Profile 4 (orange) is characterized by consistently high High–High use throughout, whereas Low– Low decreases over time.

These profiles, which reflect varying levels of risk, are labeled as "low-risk," "mid-risk," "risk-increasing," and "highrisk," based on the trend and magnitude of screen time changes.

• Profile 1 low-risk (27.2%): predominantly no use, shifting to light use.

• Profile 2 mid-risk (38.4%): shifts from No–Low to Low– Low.

• Profile 3 risk-increasing (15.3%): moves from No–Low to Low–Low, then escalates to High–High.

• Profile 4 high-risk (19.1%): continual high use.

Therefore, JLCPA uncovered dynamic screen time patterns during the study period, including the impact of the COVID-19 pandemic on these trends.

Key features influencing screen time profiles identified using multinomial logistic modeling

Multinomial logistic regression was conducted to identify the predictors distinguishing the screen time profiles across six years. The covariates included sex, age, parental education level, family income, and CBCL scores. The results revealed that the most significant influencing factors were maternal education level and sex. Children whose mothers had lower education levels were more likely to belong to higher-risk profiles. Children were significantly more likely to belong to the high-risk group (Profile 4) if they had a mother with a lower education level (6.15 times), were male (1.74 times), had higher CBCL scores (1.04 times), and were older (1.01 times). The likelihood of belonging to the Risk-increasing group (Profile 3) was higher for children whose mothers had a lower education level (2.3 times), who were male (1.94 times), and children whose fathers had high school education or less (vs. college or higher) were less likely to belong to Profile 3 (0.57 times). Children were more likely to belong to the Mid-risk group (Profile 2) if they had mothers with lower education levels (2.94 times), were male (2.35 times), and were from higherincome households (1.49 times). In contrast, children whose fathers had high school education or less (vs. college or higher) were less likely to belong to Profile 2 (0.47 times).

These results highlight the influence of socioeconomic factors, particularly parental education, and child-specific factors, such as sex and behavior (CBCL scores), on shaping children’s screen time patterns (Table 3).

DISCUSSION

Our study represents one of the first long-term prospective cohort investigations in South Korea to examine longitudinal trajectories of children’s smart device use and the factors associated with these patterns over time. The baseline assess-ment (Wave 4) was conducted between October and November 2018, with annual follow-ups extending through Wave 9. A total of 313 participants completed these waves, including Waves 6 and 7, which occurred during the COVID-19 pandemic. Over this study period, children’s screen time across multiple devices increased markedly, particularly for smartphones and PCs. The proportion of children reporting no smartphone use plummeted from 40.6% to 6.6% on weekdays and 34.8% to 8.6% on weekends. Similarly, PCs non-use declined from 75.7% to 47.9% on weekdays. Although tablet use remained comparatively stable, a slight reduction in nonusers and an increase in children using tablets for less than one hour were observed. These shifts may reflect both the widespread diffusion of digital devices over the study period and age-related developmental changes, as the cohort was followed from early school age to early adolescence. These findings underscore the need for continued investigation into the factors shaping children’s digital engagement, particularly during these developmental transitions. These findings align with a growing body of international research. A meta-analysis synthesizing data from 46 studies and over 29,000 children and adolescents reported an average 52% increase in daily screen time during the pandemic [21], reflecting changes in lifestyle precipitated by widespread public health measures. Importantly, increases were documented not only in recreational screen time but also in educational uses, indicating that digital engagement during this period was multifaceted and

driven by necessity as much as leisure.

Comparisons across pre-pandemic, pandemic, and postpandemic periods reveal that elevated screen time tended to persist beyond the peak pandemic phases [30 , 31]. This suggests that behavioral shifts toward increased screen exposure may have long-term implications rather than being transient phenomena.

However, patterns differed between weekdays and weekends— with some studies reporting higher screen time during unstructured weekend periods [17 , 32], while others found reduced weekend screen time potentially due to increased physical activity at home [1]. Our findings, in concert with previous studies such as Del Pozo-Cruz et al. [10], underscore the complexity of these temporal patterns and the need for continued investigation into the contexts shaping children's digital engagement.

To better characterize usage patterns, we employed LC analysis, revealing three distinct classes for screen time engagement on weekdays and weekends: no-use, light-use, and SMTP heavy-use. By examining joint patterns, we identified four composite LCs: No–No, No–Low, Low–Low, and High–High. These groupings encapsulate a range from minimal to heavy use across days, reflecting the heterogeneity of media behavior among children. The No–No group, characterized by minimal screen exposure, supports evidence associating limited screen time with positive cognitive and emotional development outcomes [3]. In contrast, the High–High group, characterized by persistently elevated device use—particularly smartphones— was associated with greater parent-reported behavioral and emotional problems and may also be linked to greater physical health risks [4]. These findings reinforce prior research highlighting risks of excessive screen exposure. Notably, socioeconomic factors such as parental education and household income emerged as significant determinants. Children with parents of lower educational attainment and lower income were disproportionately represented in higher screen use groups, potentially reflecting reduced parental capacity to regulate media or limited awareness about screen time risks [17 - 19]. Our findings are also consistent with evidence from other geographical regions. A recent cross-sectional study in northern India reported that excessive screen use in young children was associated with parent-related practices, such as a lack of screen time limits, screen use during feeding, and limited awareness of the developmental impact of screen exposure [33]. These findings highlight the importance of understanding children's screen use within broader familial and sociocultural contexts.

Examining longitudinal trends during the COVID-19 pandemic reveals dynamic shifts. The No–No group steadily de-clined, while the No–Low group saw a transient increase during the pandemic before stabilizing. Post-pandemic, the High–High group expanded substantially, reflecting possible permanent transformations in digital consumption habits and family routines [16 , 34 , 35]. These changes underscore the pandemic's role as a disruptor that accelerated existing trends. Current evidence suggests the relationship between screen time and well-being is not consistently linear. Moderate screen use may be neutral or even beneficial for mental health, while excessive use is linked with adverse effects [36]. The variability between weekday and weekend effects further reflects the importance of environmental and behavioral context in this association [1]..

In our study, multinomial logistic regression identified maternal education and child sex as the most robust predictors of screen time trajectory membership. Lower maternal education corresponded with markedly increased odds of belonging to riskier screen time groups, while male sex also enhanced risk, and paternal education exerted protective effects. These results echo the broader literature emphasizing the influence of parental socioeconomic status and gender socialization in shaping children's digital media behaviors [3 , 4 , 15 , 17]. Nonetheless, we acknowledge that factors such as

household income, access to technology, parental mental health, and community resources also play substantive roles [17 , 18].

Taken together, these findings affirm the multifactorial and dynamic nature of children's screen time behaviors, shaped by intricate interactions among individual, familial, and societal influences. Our study highlights marked increases in screen time across various devices, particularly smartphones and PCs, with distinct usage patterns that vary between weekdays and weekends and evolve over time, especially in response to external factors such as the COVID-19 pandemic. The identification of discrete LCs representing different screen time profiles underscores the heterogeneity of children's digital engagement and its developmental implications. Socioeconomic factors, including parental education and income, alongside parenting strategies and sex differences, play critical roles in shaping these patterns. Accordingly, developing effective interventions demands nuanced, tailored approaches attentive to these complexities and contextual factors. Furthermore, the pandemic-induced shifts in screen time trajectories reflect the need for adaptive public health strategies. Given these multifaceted influences, large-scale longitudinal research remains imperative to elucidate the long-term developmental outcomes associated with screen time behaviors and to inform evidencebased guidelines and policies aimed at promoting balanced and healthy digital media use among children.

However, this study has limitations. Firstly, screen time was measured as an aggregate without distinguishing educational from leisure use, potentially obscuring differences in risk and benefit. With the surge in online learning during the pandemic, this differentiation is particularly salient. Secondly, participant recruitment yielded a sample of parents likely highly engaged in child-rearing and research participation, which may limit generalizability to the wider population.

Future research should address these constraints by incorporating detailed screen time typologies and broader demographic samples.